HERE TECHNOLOGIES × UCHICAGO CAPSTONE

Measuring Bridge Clearance from Street-Level Imagery

Real-world dimension estimation with metric 3D reconstruction and computer vision



Bridge (red) and road (blue) found automatically; yellow = measured underside

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THE UNIVERSITY OF CHICAGO
MS IN APPLIED DATA SCIENCE

Agenda

01 **Overview**

Business problem and research question

02 **Data**

Mapillary imagery + federal bridge inventory

03 **Pipeline**

Segment, reconstruct, measure, decide

04 **Results**

Accuracy, coverage, generalization

05 **Impact**

Recommendation for HERE

Team Kely Zhang — MapAnything, pipeline v2 · Anna Chen — Depth Anything 3, SAM3 · Adaora Ezike — VGGT

Clearance data is costly to collect, but the imagery already exists

SITUATION › **COMPLICATION** › **RESOLUTION**

HERE maintains physical map attributes

- Clearance, width, height, offsets
- Critical for truck routing, navigation, HD maps

Collecting them is expensive

- LiDAR surveys and field audits
- Hard to repeat across a whole network

Measure from imagery that already exists

- Crowd-sourced Mapillary photos
- Pretrained 3D models give metric depth

Goal A low-cost, automated way to estimate clearance at network scale, and to know when to trust it

Research question: can pretrained 3D models measure clearance in metres?

“Can pretrained vision models, plus automatic target localization and geometry, produce useful clearance estimates, and under which conditions?”

Test 1 · Accuracy

Does the estimate agree with the official inventory?
Reference: FHWA NBI Item 54B (minimum vertical clearance)

Test 2 · Added value

Is it better than what HERE knows without any image?
Reference: constant regional median, 4.97 m

Scope Main feature: bridge vertical underclearance (Chicago metro) · Second test: stop-sign size

Three sources build a benchmark with official ground truth

Mapillary imagery

- Crowd-sourced sequences
- GPS, heading, capture time
- Camera lens parameters
- Captured 2015–2026

+

FHWA Bridge Inventory

- NBI 2025, Item 54B
- Minimum vertical clearance
- Underpasses, 3.5–8.5 m
- Illinois, Indiana, Wisconsin

=

Evaluation sets

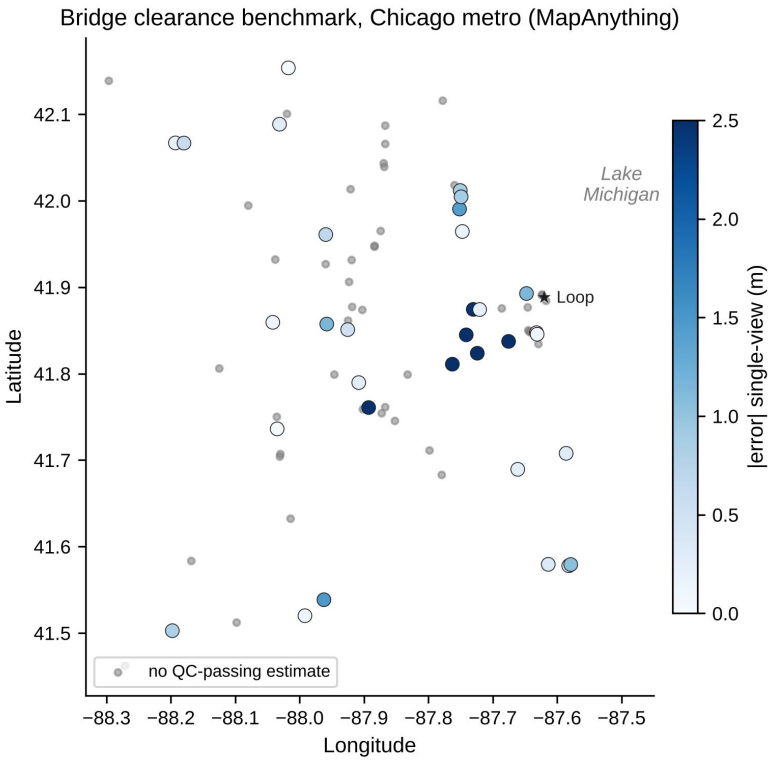
80 Chicago benchmark (development)

51 held-out Chicago bridges

120 bridges in 2 new metros

Second feature Stop signs, checked against the MUTCD standard sizes (30 in / 36 in)

80 benchmark bridges cover the full clearance range



523

candidate highway underpasses in the Chicago metro

80

sampled, stratified by clearance bin and structure type

3.9–7.9 m

NBI clearance range (median 4.97 m)

≤ 8

frames per bridge from the sequence driving toward it

Every rule and threshold was set on these 80 bridges, then frozen before any held-out test

One modular pipeline: only the 3D backbone changes

1 Retrieve
Mapillary sequence approaching the bridge

2 Segment
Bridge + road masks (Mask2Former, Vistas)

3 Reconstruct
Metric 3D points from 1 or 4 frames

4 Measure
Underside to RANSAC road plane

5 QC
Camera height, columns, distance

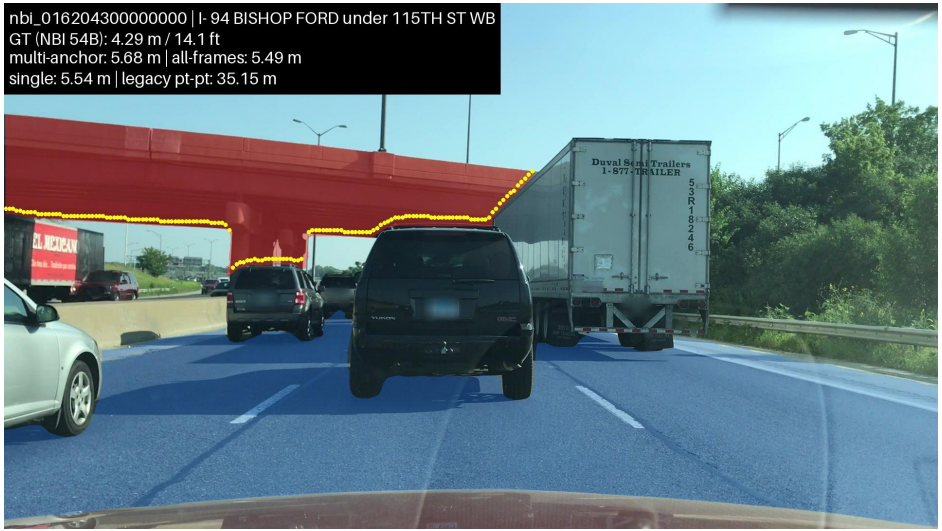
6 Decide
Accept / review / reject, GeoJSON out

Swappable backbones, same frames, masks and scoring:

MapAnything | **Depth Anything 3** | **VGGT**

≈ 6 s per bridge on one L4 GPU

Clearance = deck underside to a RANSAC-fitted road plane



I-94 Bishop Ford under 115th St · NBI 4.29 m · estimate 4.29–5.54 m across views

1. **Find the underside**

   Lowest bridge pixel with road beneath it, per column

2. **Lift to 3D**

   Look up each pixel's metric 3D point

3. **Fit the road**

   RANSAC plane through road points

4. **Measure**

   Median distance, underside to plane

Why it matters Point-to-point measure: 25.8 m error. Road plane: under 1 m.

Every estimate gets a decision, without using ground truth

Quality check

Camera 0.7–3.5 m high
≥ 10 underside columns
Bridge within 90 m

Fail → reject

›

Plausibility gate

Estimate must be 3–9 m
Below 3 m = wrong target

Fail → reject

›

Camera rule

Wide-angle, uncalibrated lens
(k1 < −0.06)

Flag → review

›

View agreement

Single- vs multi-view differ by > 1.05 m

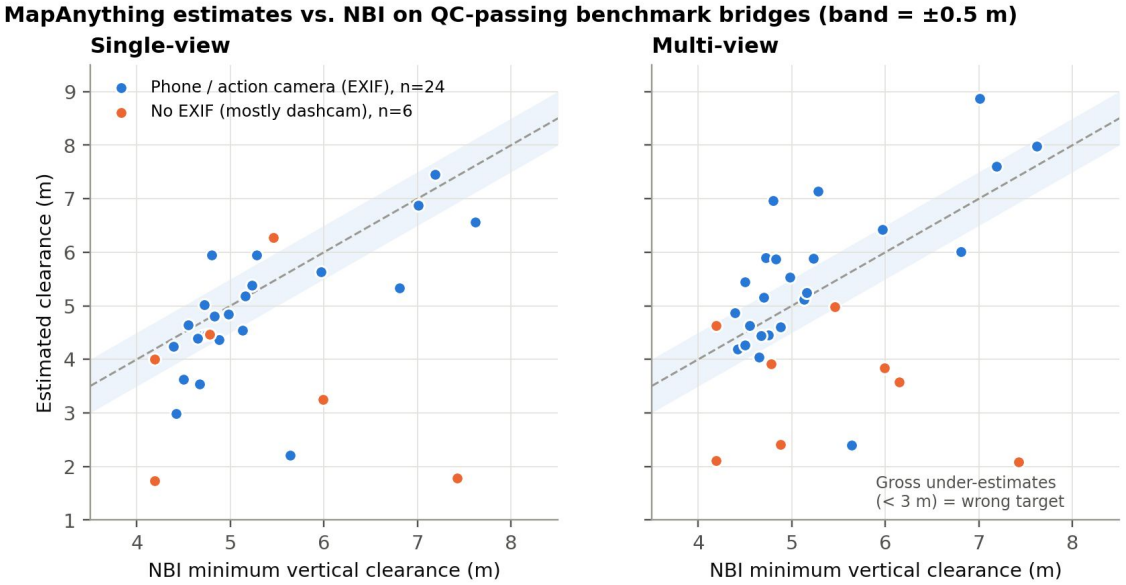
Flag → review

ACCEPT **REVIEW** **REJECT**

All thresholds set on the benchmark, then frozen

MapAnything: 0.43 m median error on quality-checked bridges

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Blue: phone / action camera · Orange: no EXIF (mostly dashcams) · Band: ±0.5 m

0.43 m

median absolute error, single-view

0.65 m

mean error on phone & action-camera imagery (0.92 m overall)

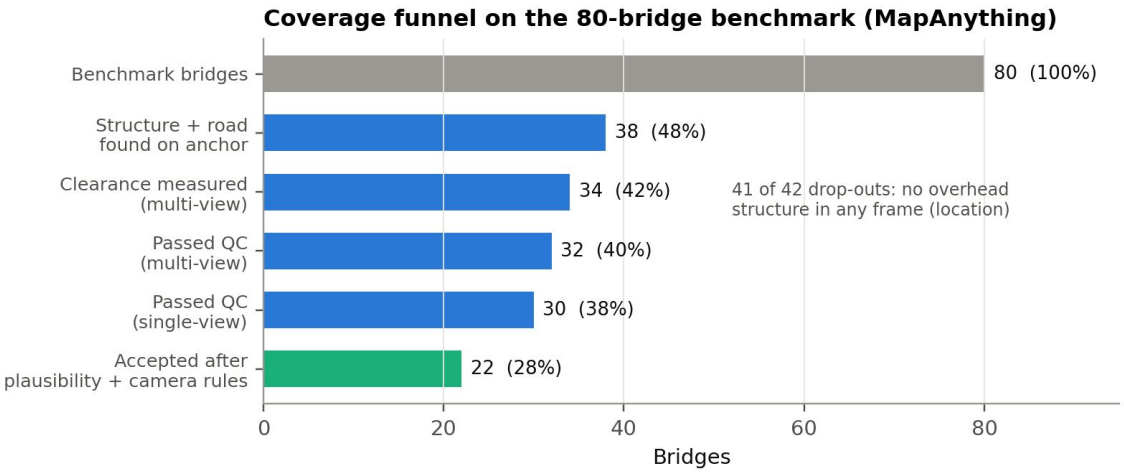
1.3 m

recovered camera height: metric scale is right

Biggest misses come from dashcams

Coverage is the bottleneck: half the bridges never get a usable frame

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41 / 80

bridges: no overhead structure in any frame

22

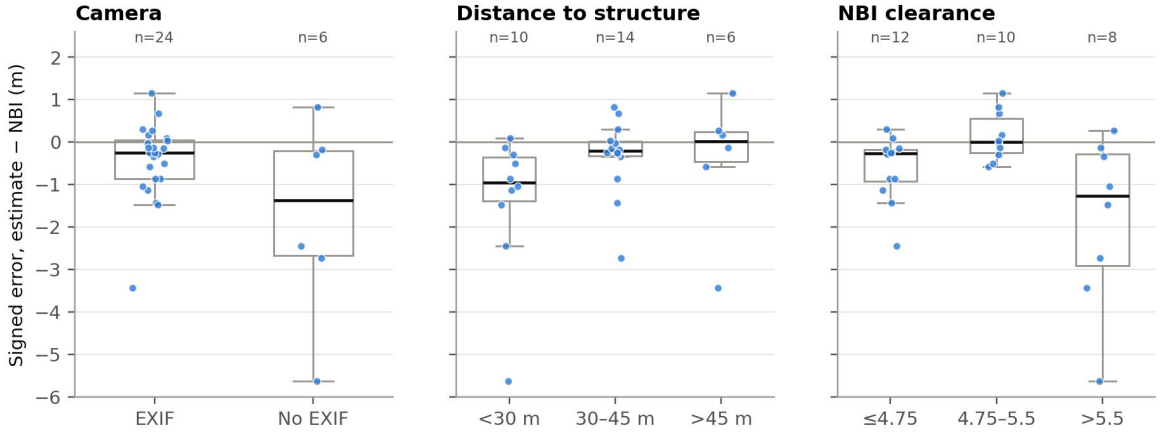
bridges (28%) accepted after all rules

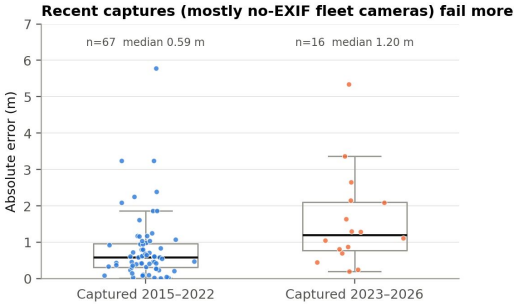
Root cause

The inventory point marks the bridge, not the road beneath it, so the nearest sequence often runs alongside or on top

Failures are predictable from metadata

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All four signals are known before comparing with ground truth, so they become rules

No-EXIF dashcams

biased low; wide-angle lens

Camera < 30 m

underestimates by 1.35 m

Estimate < 3 m

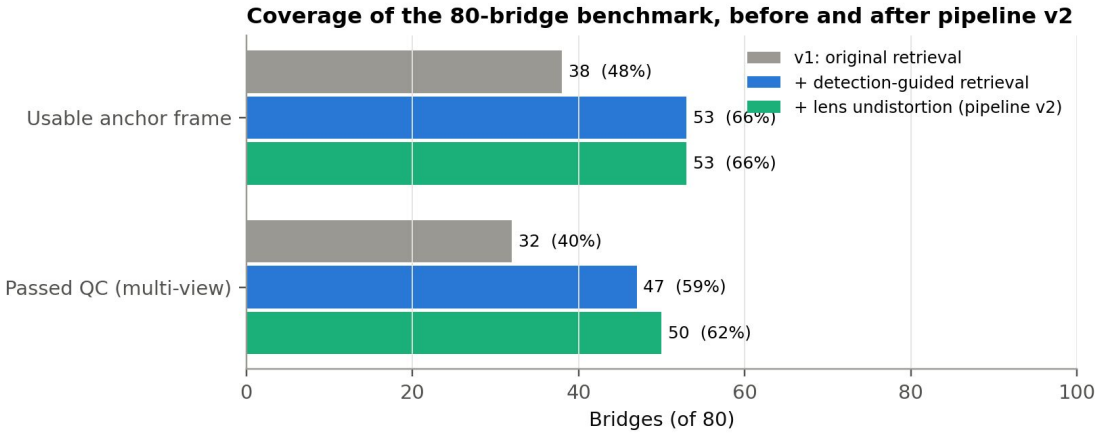
almost always the wrong target

Captured 2023+

1.20 m vs 0.59 m median error

Pipeline v2 lifts coverage from 40% to 62.5%

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+15

bridges rescued by smarter retrieval alone

+3

more pass QC after undistortion; MAE 0.92 → 0.80 m

22 → 34

bridges accepted without review

Detection-guided retrieval

Probe up to 12 nearby sequences; keep the one where the segmenter sees an underpass

Lens undistortion

Remove barrel distortion with Mapillary's k1, k2 before 3D reconstruction

Cheapest lever in the project: no change to the 3D model

Depth Anything 3: Single-View vs Multi-View

ANNA

Segmentation: Mask2Former vs SAM3

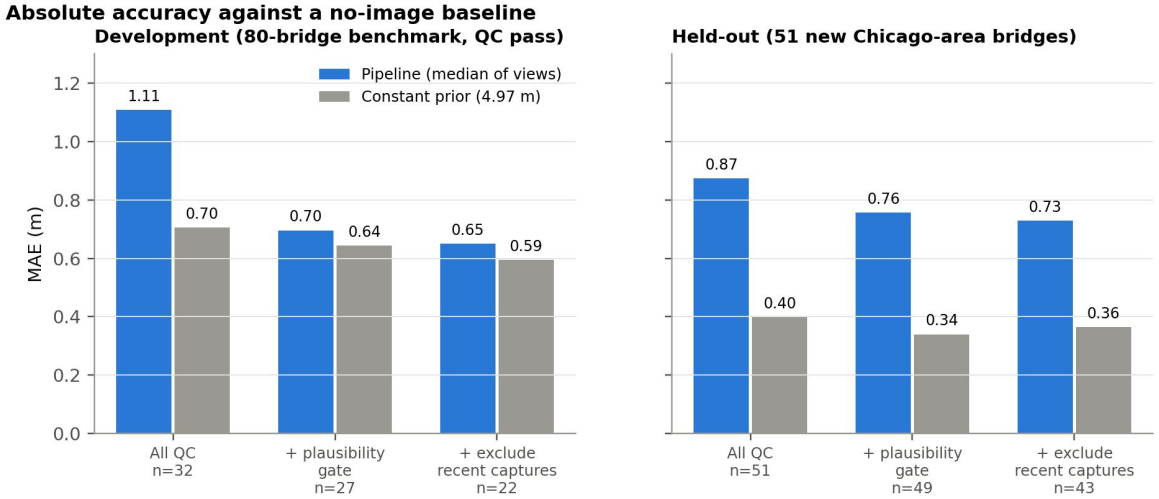
ANNA

VGGT Backbone

ADAORA

Frozen rules hold on 51 bridges never used in development

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Blue: pipeline · Grey: constant 4.97 m baseline on the same bridges

0.93 m

raw held-out error vs 0.92 m on benchmark: no overfitting

0.87 → 0.73 m

with plausibility + camera rules, 84% of estimates kept

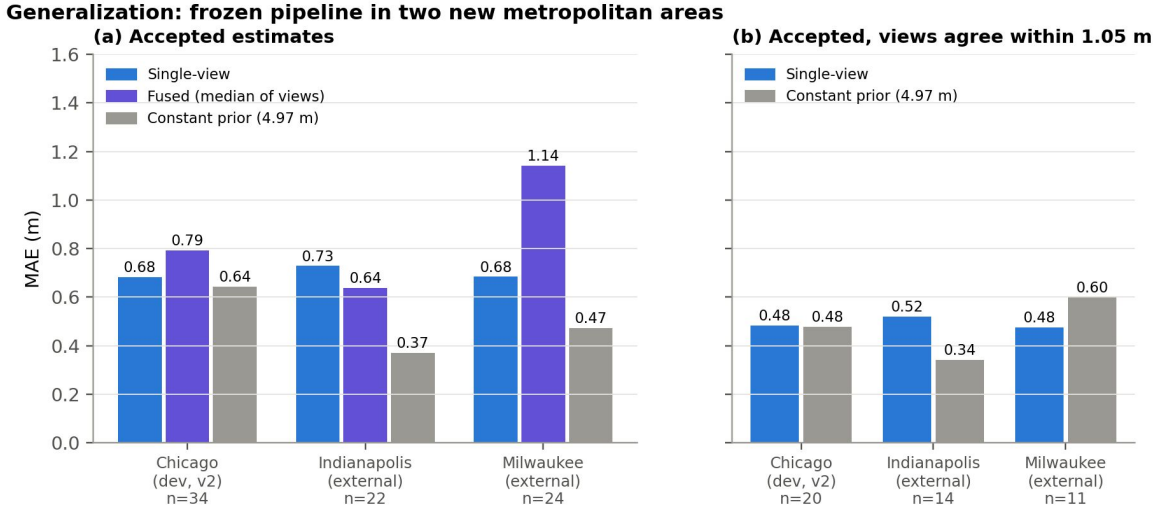
18% → 11%

share of gross errors (> 1.5 m) after the rules

+ view agreement 0.62 m error at 57% coverage

Same accuracy in two new cities, with no retuning

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120

bridges in Indianapolis and Milwaukee, never seen before

0.70 m

accepted single-view error (Chicago: 0.68 m), 80% within 1 m

3 of 4

largest errors caught by camera rule + plausibility gate

Caveat Multi-view biased +1.1 m in Milwaukee → single-view is the safer default

A constant 4.97 m is hard to beat; ranking is where imagery adds value

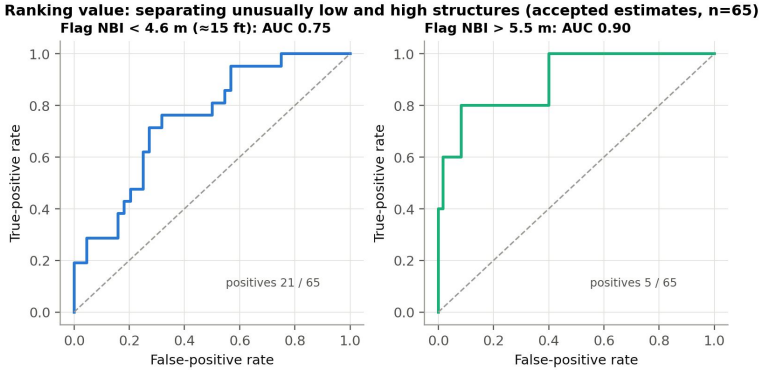
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Absolute error

Most overpasses sit at 4.3–5.2 m, so the regional median is a strong baseline. Estimates do not beat it yet.

Ranking

A constant can never find the unusually low bridges, which is what truck routing needs.



0.75

AUC, flag bridges < 4.6 m (≈ 15 ft)

0.90

AUC, flag bridges > 5.5 m

Simple screening rule

Flag any accepted estimate below 4.9 m → catches 13 of 21 truly low bridges

Recommendation: deploy as a screening and discrepancy-detection layer

1 Validate map attributes

Confident estimate disagrees with the map by > 1 m → send to analyst

Ready now

2 Screen low clearances

Rank by estimate; survey the lowest bridges first

Ready now

3 Fill missing values

Internal prioritization only; verify before publishing

After calibration

Increasing accuracy required →

≈ 6 s GPU time per bridge

Hours to screen all of Illinois

Config not a model, to add a region

Next steps: from screening to publishing values

1 Map-matched retrieval

Search along the road under each bridge with OpenStreetMap crossings (implemented)

2 Wide-angle cameras

Pass corrected intrinsics to the model; fleet cameras dominate new coverage

3 Cross-backbone ensemble

MapAnything + DA3 + VGGT; disagreement as a confidence signal

4 Calibrate at scale

Several hundred surveyed bridges per camera class before publishing

Beyond bridges Overhead sign structures · lane width · building height · stop signs already work (MAE 0.11 m vs standard size)

Thank You

Questions?

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Paper, code and results: github.com/Yichun-Zhang-ZYC

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